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Paleomagnetism of the earliest Cretaceous to early late Cretaceous sandstones, Khorat Group, Northeast Thailand: Implications for tectonic plate movement of the Indochina block

Punya Charusiri [a,*], Suvapak Imsamut [b], Zhonghai Zhuang [c], Tainpan Ampaiwan [d], Xiaoxhong Xu [c]

[a] *Earthquake and Tectonic Geology Research Unit (EATGRU), c/o Department of Geology, Faculty of Science, Chulalongkorn University, Bangkok 10330, Thailand*
[b] *Geological Survey Division, Department of Mineral Resources, Bangkok 10400, Thailand*
[c] *Chengdu Institute of Geology and Mineral Resources, Yihuan Rd., Chengdu, Sichuan, China*
[d] *PTTEP Office Building 555, Vibhavadhi-Rangsit Rd., Bangkok 10900, Thailand*

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Abstract

A total of 400 samples (33 sites) were collected from the earliest Cretaceous to early Late Cretaceous sandstones of the Khorat Group in the Indochina block for paleomagnetic study to unravel the tectonic evolution of the region. The sites were adopted from 3 traverses located in the northern edge of the Khorat Plateau, northeastern Thailand. Results indicate that almost all the sandstones exhibit similar magnetic values with an average declination (D)=31.7°, inclination (I)=30.3°, λ=59.7°, ϕ=190.9°, K=54.4, and A_{95}=3.7 at reference point 17°30′N and 103°30′E. The calculated paleolatitude points are inferred to deviate from the present latitude point by 1.2±2.3°. Only the lowermost part of the Cretaceous sandstones can pass a positive fold test at 95% confidence level. The relationship between the virtual geomagnetic poles (VGPs) of Cretaceous rocks of the Indochina plate in Thailand and those of the South China plate advocate that there is a major displacement of Indochina along the northwest-trending Red River and associated faults by about 950±150 km with a 16.0–17.0° clockwise rotation relative to the South China plate during earliest Cretaceous times. Paleomagnetic results of the early Late Cretaceous Indochina plate point to a 20–25° clockwise rotation relative to the present occurring since very Late Cretaceous (~65 Myrs)–Early Neogene times which may be due to the collision between India and Asia.
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Keywords: Khorat Group; Paleomagnetism; Indochina; Tectonics; Thailand

1. Introduction

Paleomagnetic studies of Southeast Asia have been undertaken to investigate the Mesozoic–Cenozoic (Cretaceous–Tertiary) tectonism of individual plates which were affected by the Indian–Asia collision during late Early Tertiary times (Tapponnier et al., 1986). In the South China plate, paleomagnetic data of the Cretaceous and Tertiary rocks were systematically studied by Enkin et al. (1992), Huang and Opdyke (1992), Yang (1992), and Yang et al. (2001). The Jurassic–Cretaceous and Middle–Late Tertiary (Oligocene to late Miocene) rocks of the Shan-Thai plate (or Sibumasu, Metcalfe, 1996) were paleomagnetically studied by Bunopas (1981) and later more thoroughly investigated by Ritcher et al. (1993) and McCabe et al. (1988). The Late Triassic–late Early to Middle Cretaceous and Late Miocene rocks of the Indochina plate have been studied by Maranate and Vella (1986), Achache and Courtillot (1985), Yang (1992), McCabe et al. (1988) and McCabe et al. (1993). However, up to the present, no paleomagnetic data of Middle to Late Cretaceous rocks of the Indochina plate have been obtained.

The aim of this paper is therefore to provide new paleomagnetic data of the earliest to early Late Cretaceous rocks of NE Thailand in an attempt to interpret the tectonic setting of the

* Corresponding author.
E-mail address: cpunya@chula.ac.th (P. Charusiri).

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Indochina plate in relation to the other plates since the Late Mesozoic.

2. Geologic setting

The Indochina plate which covers northeastern and eastern Thailand, Laos, Kamphuchea, parts of Malaysia, and parts of Vietnam, is bound by ancient faults and suture zones; i.e., Nan–Uttaradit suture zone in the northwest, the Red River and Song Ma suture zones in the northeast, the Sra Kaew–Chantraburi suture zone in the west and the Bentong–Raub suture zone in the southwest (Fig. 1). The upper rock units of the Indochina plate are dominated by Mesozoic continental red bed sequences (the Khorat Group), which represent the principle rock units of the Khorat Plateau, NE-Thailand. They unconformably cover deformed Paleozoic rocks. These red bed clastic rocks are

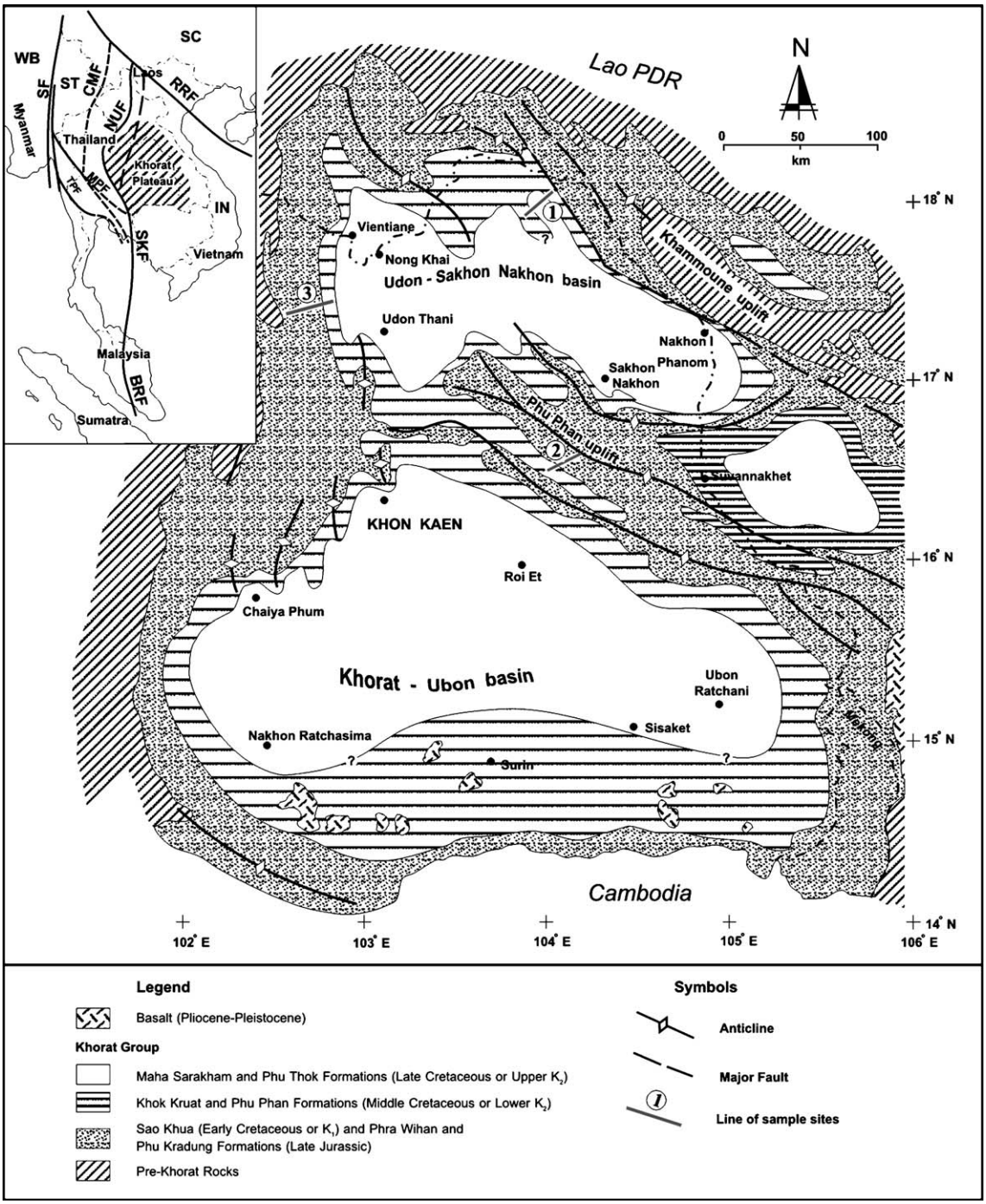


Fig. 1. Schematic geological map of the Khorat Plateau, northeast Thailand and parts of Laos (based upon our Landsat satellite image interpretation, Imsamut, 1996, Department of Mineral Resources, 2002). The inset shows the regional map of mainland Southeast Asia with the Khorat Plateau (shaded). WB = Western Burma Block; ST = Shan-Thai Block; IN = Indochina Block; SC = South China Block; SF = Sagaing Fault; CMF. = Channing–Chiang Mai Fault; SKF = Sra Kaew Chantaburi Fault; BRF = Bentong Ruab Fault; RRF. = Red River Fault; NUF = Nan–Uttaradit Fault.

believed to have been formed as molasse sequences deposited contemporaneously with collision between the Shan-Thai and Indochina plates during Triassic times (Bunopas, 1981). The Khorat Group comprises continental red and grey sandstone, siltstone and shale with some paralic and salt deposits, ranging in age from, the Late Triassic to the Cretaceous period (Ward and Bunnag, 1964).

The Khorat Group is formally divided into eight formations based on lithostratigraphy from bottom to top, these include the Huai Hin Lat, Nam Phong, Phu Kradung, Phra Wihan, Sao Khua, Phu Phan, Khok Kruat and Maha Sarakam Formations (Bunopas, 1992). Stratigraphic correlation and fossil evidence suggest that the Huai Hin Lat and Nam Phong Formations are Early Jurassic, Phu Kradung, PraWihan Formations are Late Jurassic, the Sao Khua Formation is Early Cretaceous (K1), and the Phu Phan, Khok Kruat, and Maha Sarakam Formations are Middle to Late Cretaceous (K2) (Department of Mineral Resources or DMR, 2001). Regional distribution of these formations in Thailand with an extension to Laos is shown in Fig. 1.

However, the new clastic sedimentary sequences of the Phu Thok Formation as proposed by Sattayarak (1983) have been proved magnostratigraphically to be early Late Cretaceous (K2) by Imsamut (1996). These large cross-bedding, aeolian - dominated, red clastic sequences are herein regarded as the upper K2 rocks (Fig. 1). They overlie the fluvial-dominated Khok Kruat Formation and are regarded as the upper part of the Khorat Group (Imsamut et al., 1995; Department of Mineral Resources or DMR, 2001). Rocks of the Phutok Formation were previously reported as part of the Khok Kruat Formation by Maranate and Vella (1986) and Yang and Besse (1993). However, a recent detailed drill-hole investigation by DMR in areas dominated by rocks of the Phu Tok and Mahasarakham Formations (e.g., line 1 in Fig. 1) reveals that the clastic-dominated Phutok Formation is the youngest formation of the Khorat Group. They conformably overlie the evaporite-dominated Mahasarakham Formation without a major break (DMR, 2001).

3. Previous studies

A paleomagnetic study of the Indochina Mesozoic and Cenozoic rocks was first done in Thailand by Tarling (1975), and subsequently by Barr et al. (1978), Bunopas (1981), Maranate and Vella (1986), Imsamut (1996). Achache and Courtillot (1985) made detailed paleomagnetic studies on Late Triassic sedimentary rocks from northeastern Thailand. McCabe et al. (1988) studied paleomagnetism of Late Miocene basalts from the southern part of northeastern Thailand. Yang (1992) and Yang and Besse (1993) studied the paleomagnetic record of Mesozoic rocks from the Huai Hin Lat to the Khok Kruat Formations (49 sites) in the northwest of northeastern Thailand. Van der Voo (1993) summarized the paleomagnetic studies of the Indochina plate. Richter and Fuller (1996) discussed the paleomagnetic data of Indochina and Shan-Thai in terms of extrusion tectonic model.

4. Location of samples

Three traverses (Fig. 1) in the northern part of the Khorat Plateau were selected for our paleomagnetic study. These transects were made almost perpendicular to a regional trend of the Late Cretaceous age (K2) clastic sequences where several good outcrops are well recognized. Line 1 is the reference section of the youngest K2 rock sequence—Phu Thok Formation, which is located between 103°53′E, 18°07′ N and 103°58′E, 18°14′N. Approximately 265 samples in 18 sites were collected from this traverse. Line 2 is the reference section for the early K2 rock sequence—Phu Phan Formation, located at 104°08′E, 16°29′N. About 80 Phu Phan Formation samples in 7 sites and 15 Sao Khua Formation samples in 2 sites were collected. Line 3. is the reference section of the K1 sequence—Sao Khua Formation, which is located at 102°30′ E, 17°15′N. Along this line 15 Sao Khua Formation samples from 2 sites, 10 Phu Phan Formation samples from 1 site, and 16 Khok Kruat Formation samples from 3 sites, were collected.

5. Sample collection and petrographic investigation

Samples were selectively collected at the sites with a hand-drilling machine and were oriented with a Brunton magnetic compass. The core sample was generally 2.5 cm in diameter and 8–15 cm in length, depending on rock hardness. Usually 8–10 cores were drilled per sample site, covering an area of 100 m^2. They were labeled with both the inclination and declination directions. The error of the core sample orientation is not more than 2–3°. One or two specimens (2.5 cm in diameter and 2.2 cm in length) were selected from each core sample.

Polished sections of 30 selected samples illustrate that the magnetic minerals are very fine- to fine-grained minerals (0.01–0.1 mm). They usually occur as tiny grains in sediment matrix as well as cementing material in the porous areas of sandstone or siltstone. The magnetic minerals can be separated into 2 types. These include: 1) detrital magnetic minerals (0.05–0.1 mm), as recognized by metallic lustre, detrital, anhedral grains (Fig. 2A and B), and 2) secondary magnetic minerals (<0.01 mm) which are mostly iron oxide-cement and reprecipitated as euhedral grains of hematite (Fig. 2C and D).

5.1. Detrital magnetic grains

The red bed sandstones of the Sao Khua, Khok Kruat and Phu Thok Formations are dominantly composed of detrital hematite. They are characterized by subrounded, anhedral, anisotopic, bright to yellowish white grains. Reaction rims are common due to the alteration of hematite grains to goethite. Some magnetic detrital grains are subhedral to anhedral in form and display a cubic cleavage system. They are mostly grey in color and are interpreted to be is original magnetite. Magnetite contains exsolution or oxidation lamellae of anisotopic, bright white to yellowish white hematite and ilmenite.

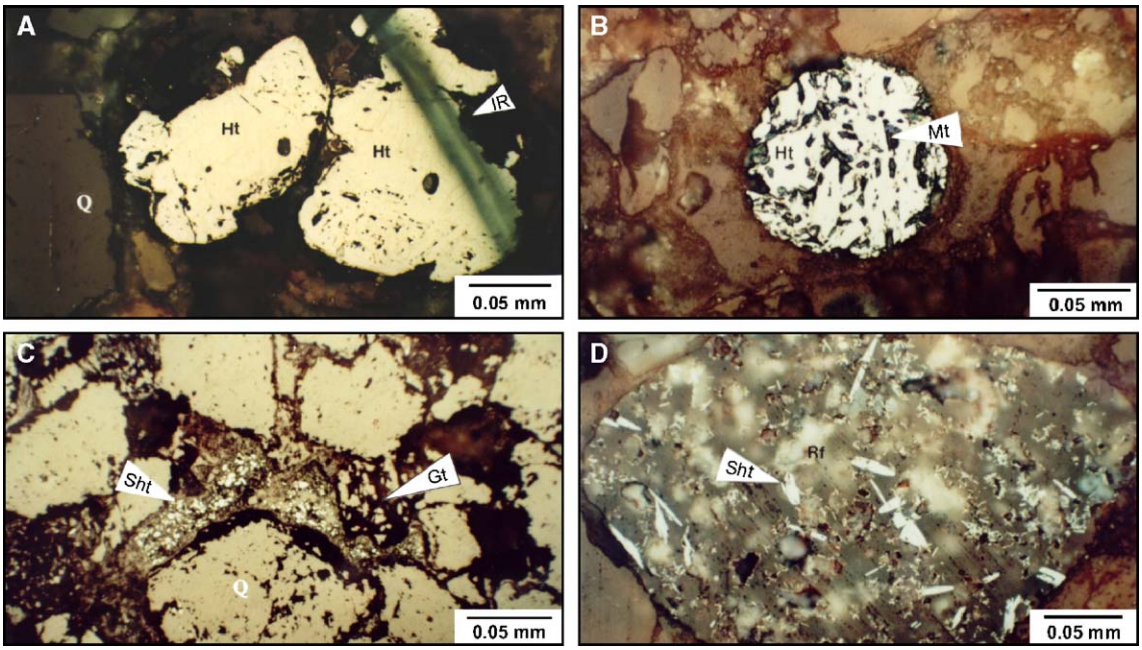


Fig. 2. Photomicrographs of red sandstones (locations in Table 1), (A) detrital hematite (Ht) grain showing internal refraction (IR) and some oxidation lamellae, bright white to yellowish-white within the hematite grain, red arkosic sandstone (site 5). The photomicrograph also shows quartz (Q) grain, (B) polished section of a fine-grained sandstone sample (site 1) showing the detrital grains of subhedral, cubic cleavage system, gray color, isotopic grain magnetite (Mt) exsoluted by anisotopic, bright white to yellowish white, hematite (Ht), (C) arkosic sample (site 5) showing the occurrence of yellowish-white secondary hematite (Sht) and goethite (Gt) at ferrugenous cement, and (D) closed-up view of a volcanic clast in quartzitic sandstone (site 30) showing randomly oriented, subhedral–euhedral, anisotopic and yellowish-white color secondary hematite (Sht) in contact with a volcanic fragment (Rt) (all photographs are under cross-nicols). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

Only the quartizitic sandstone of the Phu Phan Formation contains detrital magnetic grains with an angular shape and yellowish white hematite alteration. The other formations only contain subhedral, cubic cleavage, grey coloured, magnetite with exsolved ilmenite.

5.2. Secondary magnetic minerals

The most common secondary mineral is goethite, which occurs as iron-oxides coated around detrital sand grains or within ferrugenous cements. Some grains are reprecipitated to euhedral secondary hematite presumably at higher temperature conditions (Tarling, 1983). These secondary hematite grains (<0.01 mm) are characterized by a randomly oriented, subhedral to euhedral, hexagonal form with fibrous to lath shape, anisotopic and yellowish white.

6. Paleomagnetic treatment and analysis

Paleomagnetic work was performed at the Centre of Paleomagnetic Laboratory, Chengdu Institute of Geology and Mineral Resources, Sichuan, China. All samples were measured for natural remanent magnetization (NRM) using a Schonstedt DSM-2 spinner magnetometer. After that, they were demagnetized progressively in 9–14 steps for thermal demagnetization. The temperatures for the incremental stepwise thermal demagnetization, were progressively increased in series such as 100, 150, 200 ... to 680 °C, and 700 or 730 °C.

6.1. NRM intensities

The NRM of most types of rocks are characteristically weak, particularly in the arkosic rocks of the Khok Kruat Formation. Intensities of 1.1 to 10 mA/m are common in the Sao Khua, Khok Kruat and Phu Thok Formations and intensities of 3.1–25.0 mA/m are common in the Phu Phan Formation. Most sites, which constitute 20% of the total, were not used in generating the main pole of each site because of low magnetic intensity (less than 0.7 mA/m).

6.2. Magnetic components

The magnetic components in the rocks can be divided into 3 types. There are 1) component A; normally showing random directions and removed by 100 °C, which we suggest is due to the ambient magnetic-field, 2) component B; usually exhibiting N–S direction and removed between 100 and 250 °C (B_1) which is almost entirely carried by goethite and 250–450 °C (or up to 600 °C) (B_2) which is secondary hematite, and 3) component C; invariably NE–SW direction, removed above 450–550 °C (C_1; mainly magnetite and titanomagnetite) and above 600 °C (C_2; essentially hematite). Often all three components are present in these Cretaceous samples. After 650–730 °C, the new magnetic components from the laboratory-induced mineral were usually encountered in the opposite direction to that of the primary magnetism.

The 52 samples altogether were not used in generating the main pole of each site. This is because of the magnetic intensity

of secondary magnetization (A and B), that is more than 70% and infers unstable in magnetic intensity of rocks beyond steps of 450 °C.

7. Paleomagnetic results

7.1. Results from the Early Cretaceous (K1) rocks

Twenty arkosic sandstone and siltstone (4 sites) were selected from the Sao Khua (K2) Formation. The rocks dip at low angles (in general ~10°). The average NRM range of the Sao Khua Formation sandstone is medium (at about 2.8–5.1 mA/m). The general pattern of demagnetization is shown in Fig. 3A–D. Three components have been presented using thermal demagnetization. Component B_1 (and some component A) were completely removed below 300 °C. Component B_2 was generally removed below 450 °C. The higher component C was partly removed at step of 500–550 °C (25% NRM) and was largely removed at 650–680 °C (60% NRM).

Zijderveld plots of several samples show the trend of magnetization in N–S at steps below 450 °C (Fig. 3A–D). At higher temperatures, the magnetization from the plots changes to a NE–SW direction which is mainly consistent with the intensity plots. It clearly indicates hematite and some magnetite magnetization. However, in a few samples, the higher demagnetization (after 650 or some 600 °C) is rather noisy and misses the origin. Therefore, the middle temperature component C (450–550 °C) was selectively chosen and carefully calculated for the VGP pole from several of the Sao Khua Formation samples. Some samples display a preservation of primary remanent magnetization and represent only one high component. Intensity plots (Fig. 3A–D) show a constant or very slow gradual decreasing intensity until 550–600 °C. Then, a very sharp drop in intensity occurred and the remaining magnetization value is less than 10% NRM. Zijderveld plots of these samples indicate only one direction (NE direction) in each clean temperature.

Paleomagnetic data from the Sao Khua Formation samples after tilt correction plot on an equal area projection in the northeast quadrant and are distributed at declinations of between 24.7° and 38.5° and inclinations between 21.8° and 54.0° (Fig. 5A). When structural corrections are applied, a slightly better grouping of the directions occurs (Fig. 5B). A positive fold test at the 95% level (ks/kg = 4.7 > $F(7,7)$ = 3.79) was undertaken from samples in this formation. A pole of earliest Cretaceous is derived at λ = 59.7°N, ϕ = 178.2°E and A_{95} = 5.7°.

The average paleomagnetic direction, D = 31.8°, I = 38.3° and A_{95} = 5.7 is similar to Maranate and Vella's (1986) work (mostly normal polarity; D = 33.0°, I = 35.0° and A_{95} = 8.0 from 48 samples/16 sites) and Yang's (1992) work (D = 26.6°, I = 37.3° and A_{95} = 2.6 from 73 Phu Kradung–Sao Khua Formation samples/10 sites).

The remagnetization of the Sao Khua Formation occurred before deformation in early Tertiary shortening (Lacassin et al., 1988) and was probably acquired shortly after deposition of the sediments. However, the mean direction before and after bedding tilt corrections differs by only one degree (Table 1 and Fig. 5).

7.2. Results from the middle K2 rocks

The 54 quartzitic and arkosic sandstone samples of the Phu Phan Formation (8 sites) and the Khok Kruat Formation (3 sites) were collected. Generally the dips of these two rock formations are very gentle (5–10°).

The NRM of the Phu Phan Formation samples are moderate to high (at about 3.1–127.0 mA/m). The general pattern of demagnetization is shown in Fig. 3(E–I). The polarity of the samples observed in sites 28, 30 and 33 are not reverse, they are inferred anomalous. The general pattern (approximately 70%) suggests the strong preservation of magnetite and represents only one high-temperature component. Zijderveld plots of these samples indicate only one direction (NE) in each temperature increment. However, the higher demagnetization (after 650 or some 600 °C) is rather noisy and misses the origin. The other pattern, showing the component B_1(average 5–10% NRM), is completely removed below 250 °C. Component B_2 (average 20–40% NRM) is generally removed below 450 °C. The higher temperature component C is rapidly removed at the step of 500–550 °C (60–70% NRM) stage and is almost completely removed at 650–680 °C (5–10% NRM). Zijderveld plots of these samples (Fig. 3 E–I), showing the demagnetized pattern, are quite similar to those of the Sao Khua Formation sandstone (Fig. 3A–D). Results indicate that the appropriate demagnetized temperature was 350–550 °C and these results are selected to calculate the VGP pole of this formation.

The average NRM of the Khok Kruat Formation is relatively weak (at about 0.9–2.0 mA/m). The general pattern of demagnetization is shown in Fig. 4(A–C). No reverse polarity samples were clearly observed. The demagnetized patterns of the Khok Kruat Formation samples are similar to those of the Sao Khua Formation, but have tremendous secondary magnetization. The selected data are those of the high component C, which is rapidly removed at the 500–550 °C (20% NRM) step and is almost completely removed at 600–680 °C (25% NRM). This suggests the presence of hematite and some magnetite. The intermediate cleaned temperature of component C (400–550 °C) is selected to calculate the VGP pole of the formation.

Paleomagnetic data of the samples from the Phu Phan and Khok Kruat Formations plot in the northeast and some southwest quadrants of an equal area projection, with declinations of between 22.0° and 49.2° and inclinations between 5.9° and 54.8°. When structural corrections are applied, a slightly better grouping of the directions occurs (ks/kg = 1.1). However, a positive fold test is inconclusive because the rocks show low dip angles.

The average paleomagnetic direction, D = 31.4°, I = 27.1° and A_{95} = 9.4 slightly differs from Yang's (1992) work (D = 32.0°, I = 38.2° and A_{95} = 4.1 (550–670 °C) from 77 samples/10 sites.)

A Lower Cretaceous pole is derived for the Phu Phan Formation at λ = 62.5°N, ϕ = 199.6°E and A_{95} = 13.5° and the Khok Kruat Formation pole at λ = 58.7°N, ϕ = 178.5°E and

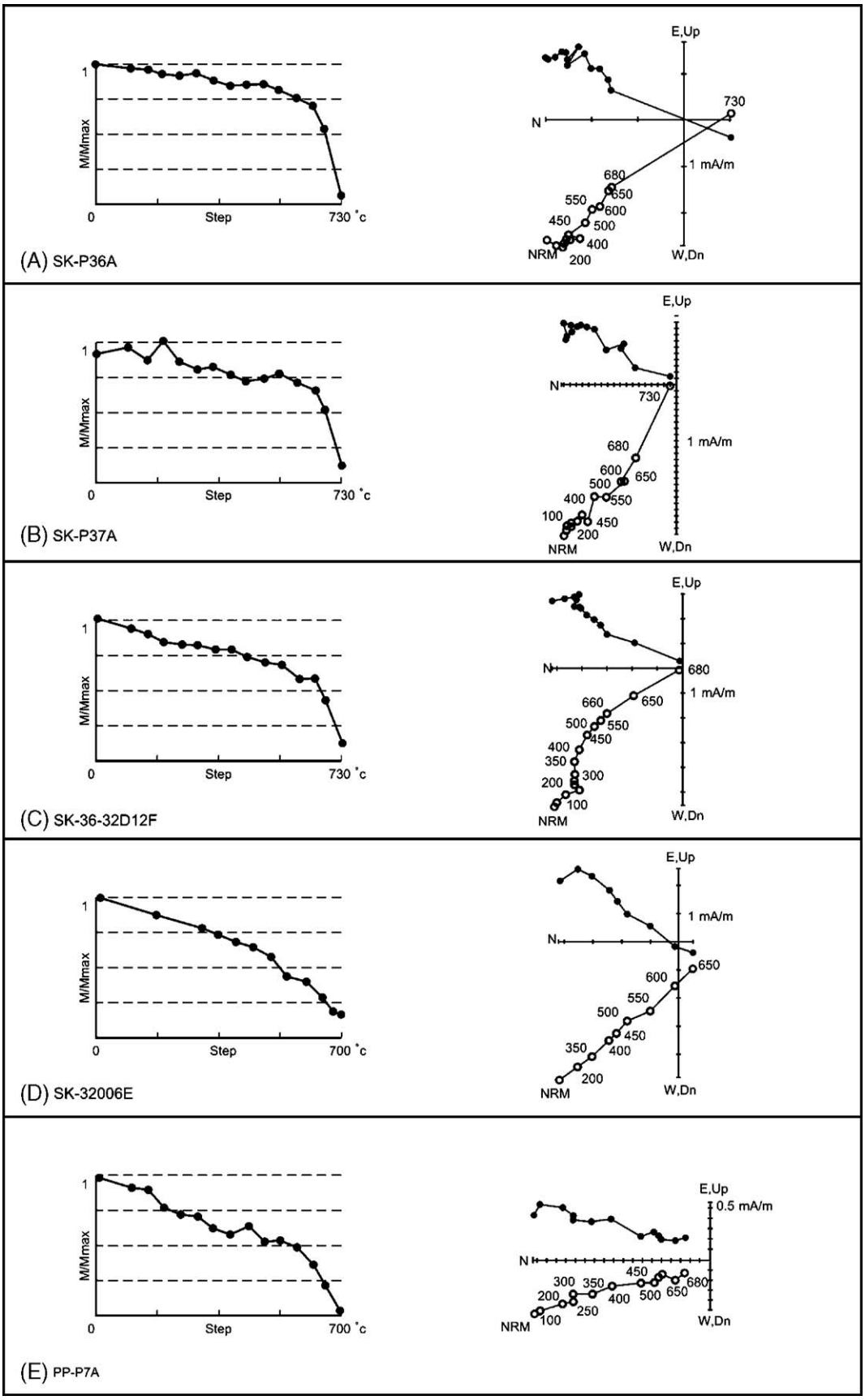


Fig. 3. Representative orthogonal (Zijderveld) plots (right) and normalized thermal demagnetization intensity curves (left) for the Early Cretaceous (K1) samples of the Sao Khua Formation (A, B, C and D) and the Middle Cretaceous (lower K2) samples of the Phu Phan Formation (E, F, G, H and I).

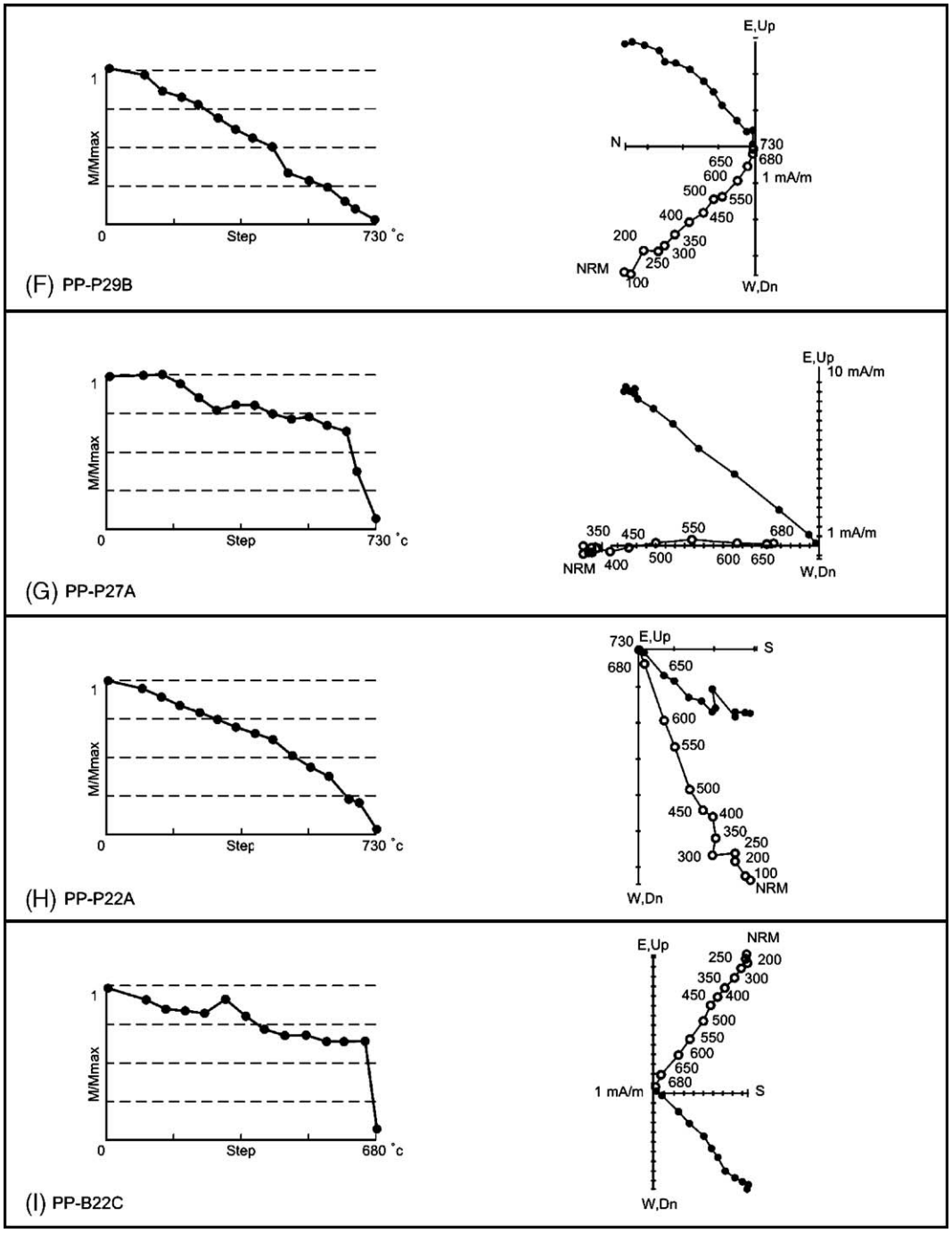


Fig. 3 (*continued*).

A_{95}=8.3°. The calculated mean of Lower Cretaceous from 8 normal sites is derived at λ=59.7°N, ϕ=192.7°E and A_{95}=9.4° (see also Table 1 and Fig. 6).

7.3. Result from the youngest K2 rocks

189 arkosic sandstone and siltstone samples of the Phu Thok Formation (18 sites) were collected from the Phu Sink–Phu Thok Range whose dips are very gentle (in general 5–10°). The average NRM are moderate (at about 1.1–55.0 mA/m). The general pattern of demagnetization is shown in Fig. 4(D–I). Results from the probably reverse polarity samples (indicated by reverse declination and normal inclination) are found at sites 5, 10, 12 and 16 and ratio of normal/reverse is about 7:1. This evidence suggests that primary magnetization data were encountered in the Phu Thok Formation samples.

At least three patterns of demagnetization are recognized in this formation. The first pattern is similar to the general pattern of the Sao Khua Formation samples, showing that the component B_1 and some component A (average 10% NRM) is completely removed below 250 °C. Component B_2 (average 30–40% NRM) are generally removed below 450 °C. The

Table 1
Characteristic remanent magnetization direction for the Cretaceous rocks of the Khorat Group (predominantly use middle temperature component (350–550 °C) and some higher temperature component (550–680 °C)

Site	Location	Strike/ dip	*N*	In situ			Tilt corrected			Pole position	
				Dg (°)	Ig (°)	Ds (°)	Is (°)	ks	A_{95}	Lat (°)	Long (°)
Sao Khua formation — Earliest Cretaceous											
19	17°15′/102°30′	306/166	6	24.7	54.0	27.0	43.4	83.1	6.3	63.7	169.7
20	17°15′/102°30′	305/10	4	31.8	43.5	32.2	35.7	262.8	4.3	59.4	182.4
25	16°29′/104°08′	141/10	5	38.5	21.8	37.4	31.6	16.2	15.5	54.2	187.4
26	16°29′/104°08′	141/10	5	32.5	33.0	29.8	42.4	62.8	7.9	60.9	172.7
Formation mean			4	32.6	38.2	31.8	38.3	149.6	5.7	59.7	178.2
Phu Phan and Khok Kruat formations — Early–Middle Cretaceous											
21_p	17°15′/102°30′	295/10	4	23.3	5.9	22.7	14.0	14.2	16.6	65.9	215.0
22_k	17°15′/102°30′	295/4	3	21.5	44.7	21.7	44.7	18.2	19.0	68.2	167.7
23_k	17°15′/102°30′	300/4	3	36.2	38.9	36.7	35.1	16.4	19.9	54.9	181.0
24_k	17°15′/102°30′	300/4	3	39.0	37.7	38.6	33.8	68.9	9.7	53.2	182.4
27_p	16°29′/104°08′	130/10	5	49.2	15.0	49.5	24.9	9.8	15.8	42.2	191.9
28_p*	16°29′/104°08′	130/10	7	39.7	−42.2	40.4	−31.5	6.4	20.9	−37.8	53.1
29_p	16°29′/104°08′	136/11	8	26.6	16.0	24.6	26.2	19.6	10.5	66.1	197.8
30_p*	16°29′/104°08′	136/11	5	174.7	54.8	183.8	48.0	27.8	11.9	44.3	100.1
31_p	16°29′/104°08′	138/3	9	22.5	39.9	20.8	43.9	30.8	9.5	68.6	166.0
32_p	16°29′/104°08′	141/10	7	22.0	17.2	22.9	8.0	9.2	23.1	64.3	221.4
33_p*	16°29′/104°08′	141/10	5	237.7	−43.0	240.2	−51.4	77.1	7.1	−33.6	13.2
Normal sites			8	32.9	24.2	31.4	27.1	27.6	9.4	59.7	192.7
Reverse sites			3	30.2	−50.7	34.4	−41.3	12.8	27.6		
Formation mean			8	32.9	24.2	31.4	27.1	In situ	9.4	59.7	192.7
Phu Thok formation — Middle–early Late Cretaceous											
1	18°07′/103°53′	332/5	9	47.1	32.5	47.8	27.7	48.1	6.7	44.0	188.3
2	18°07′/103°53′	320/5	11	20.6	30.3	22.0	26.0	23.9	8.6	68.6	197.6
3	18°07′/103°53′	332/7	16	32.0	32.4	33.7	26.5	15.0	9.0	57.4	192.8
4	18°07′/103°53′	332/7	13	32.2	33.5	34.2	27.4	26.1	7.6	57.0	191.6
5*	18°07′/103°53′	347/7	5	243.4	47.2	241.2	53.9	4.4	29.8	12.7	55.2
6	18°07′/103°53′	347/7	9	34.1	30.3	36.5	25.5	12.7	13.1	54.6	193.0
7	18°07′/103°53′	332/6	19	27.6	30.9	29.4	25.9	18.8	7.4	61.5	194.7
8	18°07′/103°53′	010/9	9	25.1	31.0	29.0	25.6	14.8	12.1	61.9	195.3
9	18°07′/103°53′	005/5	13	27.1	35.2	29.5	31.8	15.1	10.0	61.8	187.2
10*	18°07′/103°53′	005/5	4	184.3	19.2	182.9	19.9	11.1	21.0	63.1	289.4
11	18°07′/103°53′	005/5	15	42.2	28.4	43.5	24.3	10.7	11.1	47.8	192.2
12*	18°14′/103°58′	144/11	5	215.1	−5.8	215.7	−14.2	2.3	41.1	−53.8	182.5
13	18°14′/103°58′	144/11	15	34.1	27.8	31.5	38.1	12.3	10.3	59.9	178.7
14	18°14′/103°58′	145/7	13	27.1	25.9	25.2	32.1	11.6	11.4	65.8	187.2
15	18°14′/103° 58′	145/7	8	26.9	23.6	25.0	29.9	17.1	12.0	66.0	190.5
16*	18°14′/103°58′	145/7	5	237.4	47.0	236.8	39.4	5.2	30.8	22.0	46.3
17	18°14′/103°58′	150/10	10	24.3	23.4	21.2	31.2	15.5	11.2	69.7	189.0
18	18°14′/103°58′	150/10	10	36.4	19.2	35.2	26.8	16.1	11.0	56.0	192.0
Normal sites			14	31.2	29.1	31.8	28.7	111.4	3.5	59.4	190.8
Reverse sites			4	216.9	29.1	216.1	27.0	4.8	31.8		
Formation mean			14	31.2	29.1	31.8	28.7	111.4	3.5	59.4	190.8
Mean of earliest to Middle Cretaceous rocks			*26*	–	–	*31.7*	*30.3*	*54.4*	*3.7*	*59.7*	*190.9*

Remark: *N* = number of samples or sites in calculation for the mean, Dg = declination before tilt correction, Ig = inclination before tilt correction, Ds = declination after tilt correction, Is = inclination after tilt correction, kg = precision parameter (before tilting), ks = precision parameter (after tilting), A_{95} = 95% cone of confidence about mean.

*Excluded from calculation of the formation mean.

$_p$ = Samples of Phu Phan formation, $_k$ = samples of Khok Kruat formation.

Fold test.

Mean of Earliest Cretaceous ks/kg =4.7>F(7,7)=3.79 indicating positive fold test at 95% confidence level.

Mean of Early–Middle Cretaceous ks/kg=1.1.

Mean of Middle–early Late Cretaceous ks/kg=1.01.

higher temperature component C is rapidly removed at the 500–550 °C (25% NRM) step and is almost entirely removed at 650–680 °C (40% NRM). This clearly indicates the occurrence of hematite and some magnetite grains. A second pattern, representing only one or two components is observed. Intensity plots exhibit the constant or very slowly gradual decreasing intensity to 550–600 °C. Then, a very sharp drop in intensity is encountered with the remaining magnetization is less than 10%

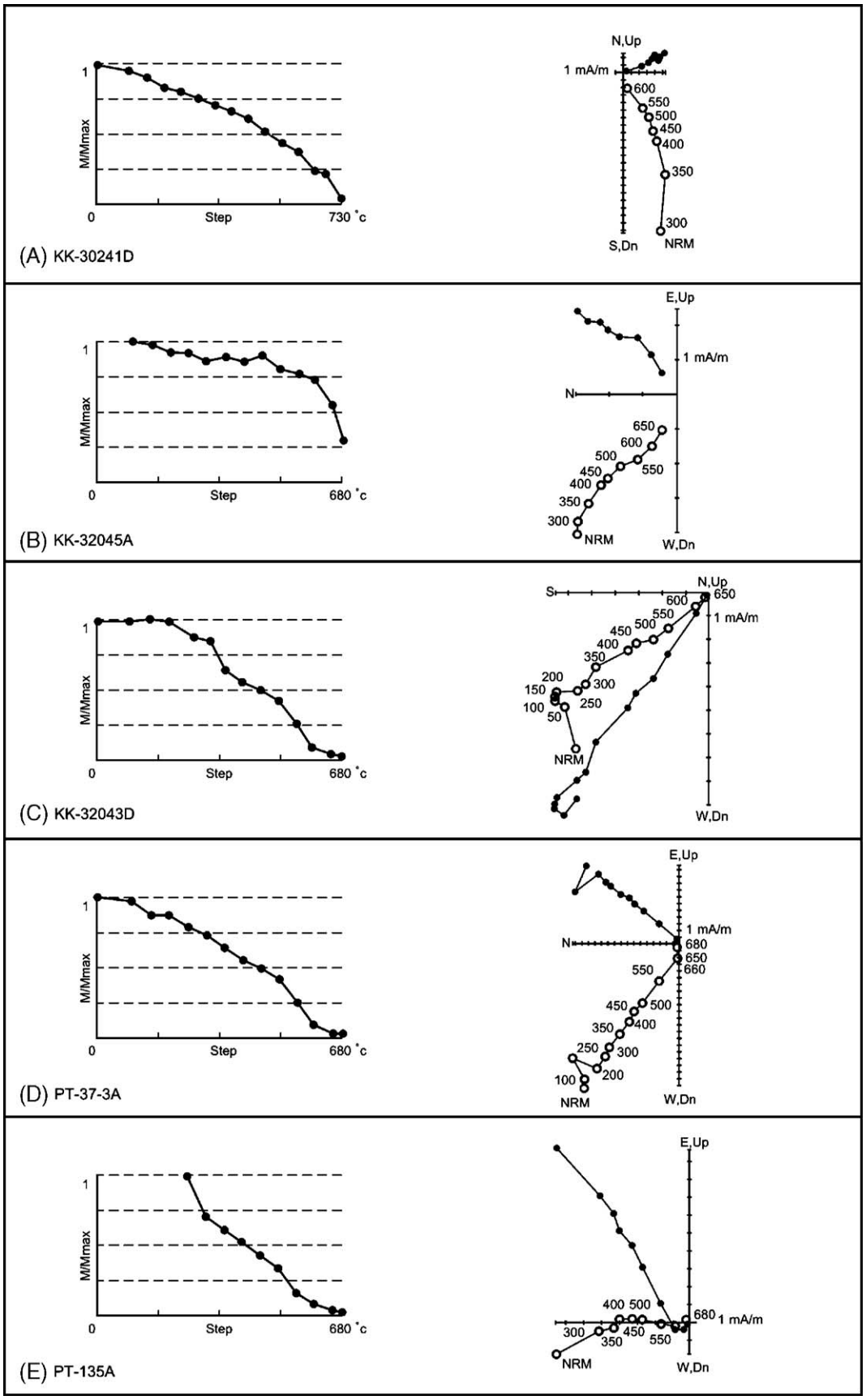


Fig. 4. Representative orthogonal (Zijderveld) plots of normalized thermal demagnetization intensity curves for the Middle Cretaceous (lower K2) samples of the Khok Kruat Formation (A, B and C) and the Late Cretaceous (upper K2) samples of the Phu Thok Formation (D, E, F, G, H and I).

NRM. Zijderveld plots of these samples represent only the NE-direction at each clean temperature. It is considered to show hematite magnetization.

The other pattern, at a demagnetized temperature range of 100–250 °C, displays a slow gradual decreasing magnetic intensity. This is followed by a rapid decrease in intensity until the 550 °C step with 10% NRM remaining. The Zijderveld plot of these samples shows a constant direction (i.e., NE-direction) at each temperature. It indicates magnetite magnetization with lesser degrees of secondary magnetization.

Paleomagnetic data of the Phu Thok Formation samples dominantly plot in the northeast quadrant of an equal area plot with declinations between 20.6° and 47.1° and inclinations of between 19.2° and 47.2°. Some reverse magnetization sites plot in the southwest quadrant at directions of between 184.3° and 243.4° but differ in inclination. When structural corrections are applied, a slightly better grouping of the directions occurs (ks/kg=1.01). However, the positive fold test is inconclusive because of the low dips of the rocks.

The mean directions of the sites before and after bedding corrections are presented in Table 1 and are shown in the equal-area projections in Fig. 5. A pole of Middle to early Late Cretaceous samples is derived from 10 normal polarity sites at λ=59.4°N, ϕ=190.8°E and A_{95}=3.5°. It is very close to the Early to Middle Cretaceous poles.

The computed overall data of earliest to early Late Cretaceous rocks in the Khorat basin has a mean of D=31.7°, I=30.3°, λ=59.4°N, ϕ=190.9°E, K=54.4 and A_{95}=3.7°.

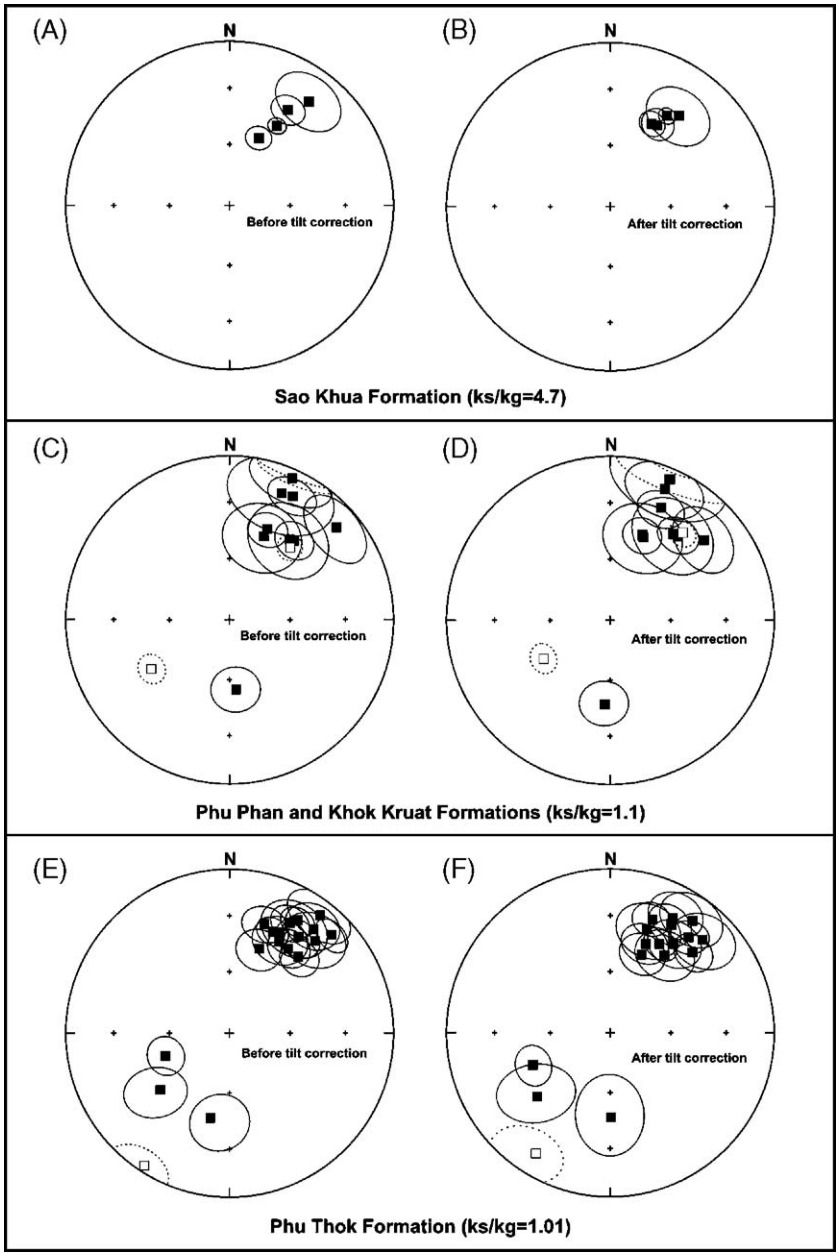


Fig. 5. Equal-area projection of site-mean direction from the in situ (left) and tilt corrected (right) of the Sao Khua (K1) Formation (A, B), the Phu Phan (K2) and Khok Kruat (K2)Formations (C, D), and the Phu Thok (K2) Formation (E, F).

8. Discussion

8.1. Hiatus in the earliest–Early Cretaceous

Several pieces of evidence advocate the appearance of the hiatus or minor unconformity between these Khorat Group sequences. Mouret (1994) suggested the presence of an unconformity between the Nam Phong and Phu Kradung Formations (Early to Middle Jurassic) that was consistent with the palynological data of Racey et al. (1994). The paleomagnetic study by Yang (1992) also suggested a similar result by indicating that the mean pole from the Triassic–Jurassic boundary (Huai Hin Lat and Nam Phong Formation) strongly contrasted with that of Jurassic–Cretaceous poles (Phu Kradung, Phra Wihan, Sao Khua–Jurassic, and Phu Phan and Khok Kruat–Cretaceous; the age inferred in his paper). This strongly implied a significant tectonic movement at the end of the Triassic. Such event is clearly supported by the presence of voluminous, north-trending, Triassic magmatism along the western edge of the Khorat plateau (Charusiri et al., 1993). Additionally, Yang and Besse (1993) also pointed out that remagnetization of Upper Triassic/Lower Jurassic sediments took place prior to the deposition of the Upper Jurassic/Lower Cretaceous sediments.

The minor unconformity is between the Phu Kradung and Phra Wihan Formations is based on well-log and seismic data in Phu Phra-1 (Mouret, 1994). Magnetostratigraphical data from Maranate and Vella (1986) indicate that the boundary between the Late Jurassic Sao Khua Formation and the Early Cretaceous Phu Phan Formation (the age referred to in their paper) could represent a large hiatus of up to 30 Myrs. This is also advocated by the abrupt difference in sediment types and paleocurrent directions (Murray et al., 1993).

The unconformable contact between the Khok Kruat and the Maha Sarakam Formations is well imaged in seismic data from oil companies (Murray et al., 1993) and drill core data for potash/rock salt exploration in northeast Thailand (Hite and Javanaphetr, 1979 and Sattayarak et al., 1991) and Laos (Smith et al., 1996). This angular unconformity indicates non-deposition or extensive erosion during the Middle Cretaceous times. Smith et al. (1996) also recognized a phase of Middle–Late Cretaceous granitic intrusion in Laos.

Our paleomagnetic results point to a similarity in D, I, λ and Φ values' data of from the Sao Khua (earliest Cretaceous), the Phu Phan and Khok Kruat (Early to Middle Cretaceous) and the Phu Thok (Middle to early Late Cretaceous) Formations. The declinations from 31.4° to 31.8°. Earliest Cretaceous inclination (38.3°) differ slightly from younger Cretaceous results (27.1–28.7°). The verified geomagnetic poles (VGPs) of these formations are quite similar in latitudes (59.4–59.7°N) but differ in longitude (178.2°E in earliest Cretaceous age and 192.7°E in younger Cretaceous rocks). The earliest Cretaceous strata exhibits a positive fold test at 95% confidence level [ks/kg=4.7>$F(7,7)$= 3.79] whereas the Early to Middle Cretaceous strata have inconclusive fold tests. We calculated their latitude differences and found that the earliest Cretaceous paleolatitude differs at −4.6±0.4°N from the present whereas the younger Cretaceous rocks differ at 2.6±0.8°N (Early–Middle Cretaceous) and 2.0±0.2°N.

The paleomagnetic results from the Early to Middle Cretaceous and Middle to early Late Cretaceous rock formations are similar in D, I, λ, Φ values and latitude differences. In this case, we cannot interpret the conformity of rocks through our paleomagnetic studies because of the lack of conclusive fold tests and the same magnetization possibly acquired in these rocks.

8.2. Time of rotations

Many geological studies of the Indochina plate concluded that large tectonic movement correspond with the beginning of the Himalayan-related deformation during the Late Cretaceous to Paleocene times, culminating in the Eocene (Mouret, 1994). During this time, the tectonic uplift of Khorat Plateau (which is in a NE–SW paleomagnetic direction), long wavelength of folds and erosion of about 3000–3500 m of the red bed sequences occurred. Subsequently, the younger Pliocene–Pleistocene basalts with N–S paleomagnetic declinations were extruded, unconformably covering the older rocks.

The previous paleomagnetic data (see Table 2) demonstrated a NE-trending paleomagnetic direction preserved in the Khorat Group from the Late Triassic to Early–Middle Cretaceous time (~230–100 Myrs). Our Late Cretaceous paleomagnetic declination results (D=31.8, A_{95}=3.5) have a similar declination to those of the Earliest–Middle Cretaceous (D=31.4–31.8°) rocks, indicating more precisely the timing of rotation of the Indochina block.

8.3. Paleoposition of the Khorat plateau (Indochina block)

The mean paleomagnetic pole of the earliest to early Late Cretaceous Indochina rocks is 59.7°N, 190.9°E (with K=54.4, A_{95}=3.7°, dP=2.3° and dM=4.1°), indicating very good and reliable results. The average paleolatitude of the formations is 16.3±2.3°N (with A_{95}=4.4°), implying that it is located at approximately the same latitude as the present site. The preserved NE direction of the rocks has a mean of about 31.7°. This clockwise rotation of the Khorat basin must have occurred after the Late Cretaceous, either as a result of Cenozoic tectonics or extensive weathering of the rocks.

When our results are compared with those of earlier workers, it is visualized that there was a slight change in paleomagnetic values from the Jurassic to Cretaceous times (Table 2). The declination angles deviated only by 15° during that period of time.

8.4. Paleoposition of the Khorat plateau relative to the South China block

Jurassic–Cretaceous paleomagnetic poles from the South China plate were produced by several geologists, they are summarized in Table 2 and Fig. 6. Pole-latitude data for Early–Middle Jurassic rocks of several microplates within the South

Table 2
List of paleomagnetic data for Indochina, Shan-Thai and South China

Formation	Age	*n*	Ds	Is	*k*	a_{95}	Pole lat.	Pole long.	References
Indochina block									
(Upper Triassic to Recent)									
Basalt, east of Thailand	Tu-Q	53	4.0	–	15.0	5.0	86.0	171.0	McCabe et al. (1988)
Phu Thok	Km-Ku	189	31.8	28.7	111.4	3.5	59.4	190.8	this study
Khok Kruat	Km	21	47.0	29.0	19.0	14.0	45.0	189.0	Maranate and Vella (1986)
Phu Phan-Khok Kruat	Kl-Km	42	31.4	27.1	27.6	9.4	59.7	192.7	this study
Phu Phan-Khok Kruat	Kl-Km	77	32.0	40.5	390.6	2.4	62.7	173.3	Yang (1992)
Phu Phan	Kl	9	45.0	19.0	10.0	41.0	46.0	197.0	Maranate and Vella (1986)
Sao Khua	Kl	20	31.8	38.3	149.6	5.7	59.7	178.2	this study
Sao Khua	Kl	48	33.0	35.0	23.0	8.0	59.0	184.0	Maranate and Vella (1986)
Phra Wihan	Ju-Kl	42	31.0	41.0	10.0	13.0	60.0	174.0	Maranate and Vella (1986)
Phu Kradung	Ju	45	38.0	29.0	21.0	7.0	58.0	188.0	Maranate and Vella (1986)
Phu Kradung-Sao Khua	Ju-Kl	73	26.6	37.3	443.0	2.3	64.0	178.1	Yang (1992)
Nam Phong	Tru-Jl	61	37.2	40.1	71.6	6.6	54.4	175.6	Yang (1992)
Huai Hin Lat	Tru	56	39.5	44.4	85.0	8.3	52.1	169.8	Yang (1992)
Khorak	Kl	10	28.1	40.5	2.4	2.4	62.7	173.3	Yang and Besse (1993)
South China block and adjacent area									
(Upper Triassic–Cretaceous)									
South China	Ku	23*	–	–	–	3.2	74.5	229.0	Enkin et al. (1991)
Motoushan fm., Chuxiong	Ku	57	45.6	46.6	12.4	14.3	–	–	Funahara et al. (1992)
Fexianguan area	Ku	189	15.0	30.8	38.0	6.1	70.8	234.0	Enkin (1990)
APWP mean South China	Km-Ku	?	–	–	–	14.2	77.9	203.8	Enkin et al. (1992)
Nanjing	Km-Ku	?	–	–	42.0	8.0	76.0	173.0	Kent et al. (1987)
Nanxing fm., Xiaguan	Km	44	6.9	47.7	36.9	8.6	83.6	152.7	Huang and Opdyke (1993)
Mangang fm., Jinggu	Km	39	79.4	43.3	38.3	9.1	18.9	170.0	Huang and Opdyke (1993)
Mangang fm, Mengla	Km	55	60.8	37.8	41.4	7.6	33.7	179.3	Huang and Opdyke (1993)
Jiangsu, Hubei	Kl-Ku	?	–	–	38.0	7.0	73.0	180.0	Kent et al. (1987)
Ya'an. Sichuan	Kl-Ku	?	2.1	–	21.7	11.3	76.3	85.5	Otufuji et al. (1990)
APWP mean South China	Kl-Km	?	–	–	–	7.0	77.8	208.4	Enkin et al. (1992)
Lhasa Terrane	K	43	3.3	29.0	11.7	28.0	71.8	70.8	Otufuji et al. (1990)
Qiangtan Terrene	K	3*	46.7	49.7	–	12.6	48.6	172.9	Otufuji et al. (1990)
West Sichuan	K	3*	0.3	26.2	–	5.1	78.8	280.0	Otufuji et al. (1990)
Eastern Yangtze block	K	5*	19.1	38.3	–	7.2	77.0	200.0	Chan (1991)
Chuxiong, west Yunnan	K	129	44.6	41.3	9.8	10.7	49.2	178.0	Enkin (1990)
Puchanghe fm., Chuxiong	Kl	72	43.7	36.2	8.1	10.7	–	–	Enkin (1990)
Xinjin area	Kl	305	18.0	36.4	56.7	4.1	71.0	219.8	Enkin (1990)
Song Da area, Yen Chau, Vietnam	Ku	8	3.2	26.7	27.2	12.9	83.2	255.6	Takemoto et al. (2005)
Song Da area, Yen Chau, Vietnam	Ku	5	12.2	40.1	263.9	4.7	78.7	188.0	Takemoto et al. (2005)
APWP Mean South China	Ju-Kl	?	–	–	–	13.5	72.3	208.6	Enkin et al. (1992)
Guanyin fm.,	Ju	62	17.4	34.8	33.3	19.2	70.9	223.5	Enkin (1990)
Bazhulu fm, Weishan	Ju	20	7.3	25.3	10.4	54.9	76.3	250.0	Huang and Opdyke (1993)
Hepingxing fm., Jinggu	Jm	49	83.3	36.8	81.5	5.4	14.0	173.6	Huang and Opdyke (1993)
Mengga fm., Luxi	Jm	24	60.9	32.0	132.3	5.9	33.2	182.8	Huang and Opdyke (1993)
APWP Mean South China	Jm		–		–	15.0	65.0	186.2	Enkin et al. (1992)
APWP Mean South China	Jl		–		–	7.4	69.4	192.4	Enkin et al. (1992)
APWP Mean South China	Tru		–		–	16.8	45.1	194.4	Enkin et al. (1992)
Shan-Thai									
(Jurassic to Recent)									
Volcanic, Shaybe-Monwya	Tu-Q	20	4.9	27.9	85.3	10.0	86.3	307.4	Ritcher et al. (1993)
Phetchaburi basin	Tm	16	22.4	10.8	26.4	24.5	66.6	206.8	Ritcher et al. (1993)
Western Thailand (1)	J	19	35.6	38.8	31.7	12.0	56.3	176.6	Barr et al. (1978)
Western Thailand (2)	J	5	34.2	32.8	11.8	9.6	57.5	184.0	Barr et al. (1978)
Kalaw red bed	J-K	80	44.7	23.4	47.1	6.1	46.4	190.6	Ritcher et al. (1993)
Yunlong	Km	20	40.2	49.9	3.9	4.4	54.6	171.3	Sato et al. (1999)
Yunlong	Km	29	40.2	49.9	3.9	4.0	56.7	180.1	Yang et al. (2001)
Yongping	Kl	12	42.0	51.5	15.7	20.6	50.8	167.3	Funahara et al. (1993)
Jinggu	Km	8	79.4	43.3	9.1	8.9	18.8	170.1	Huang and Opdyke (1993)
Mengla	Km	10	60.8	37.8	7.6	8.2	33.7	179.3	Huang and Opdyke (1993)

Remark: Q = Quarternary, Tu = Upper Tertiary, Tm = Middle Tertiary, K = Cretaceous, Ku = Upper Cretaceous, Km = Middle Cretaceous Kl = Lower Cretaceous, J = Jurassic, Ju = Upper Jurassic, Jm = Middle Jurassic, Jl = Lower Jurassic, Tru = Upper Triassic, n = number of samples, Ds = corrected declination, Is = corrected inclination, Pole lat. = VGP latitude, Pole long. = VGP longitude, K = precision parameter, A_{95} = cone of confidence, and ∗ = number of sites.

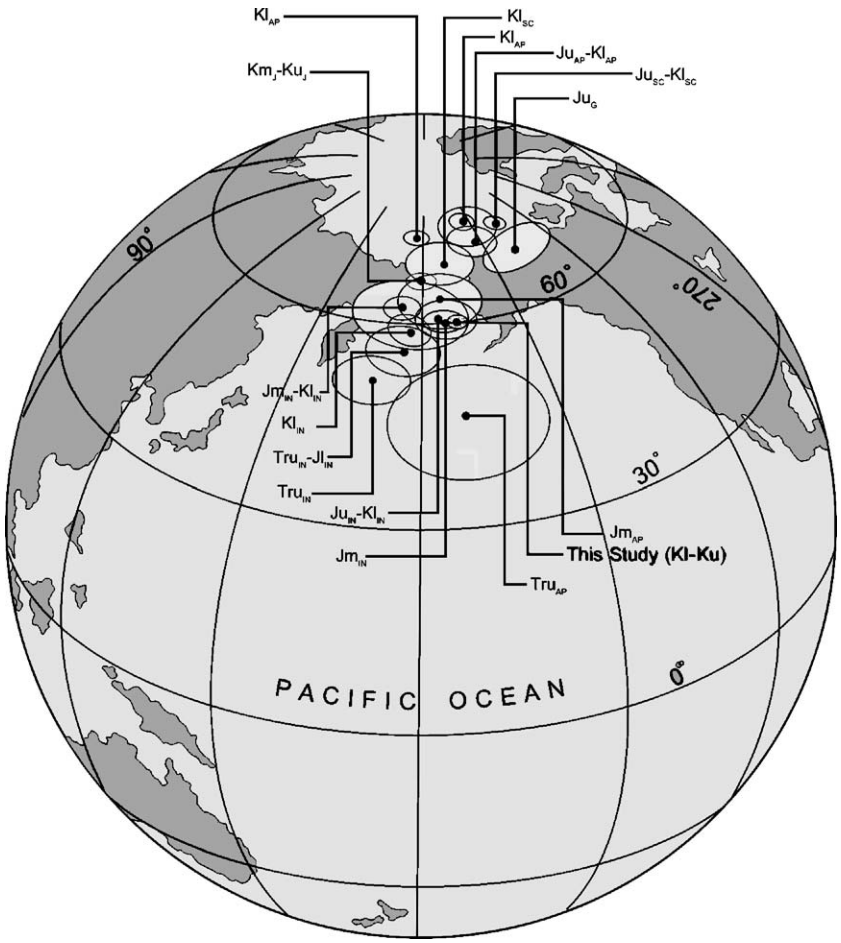


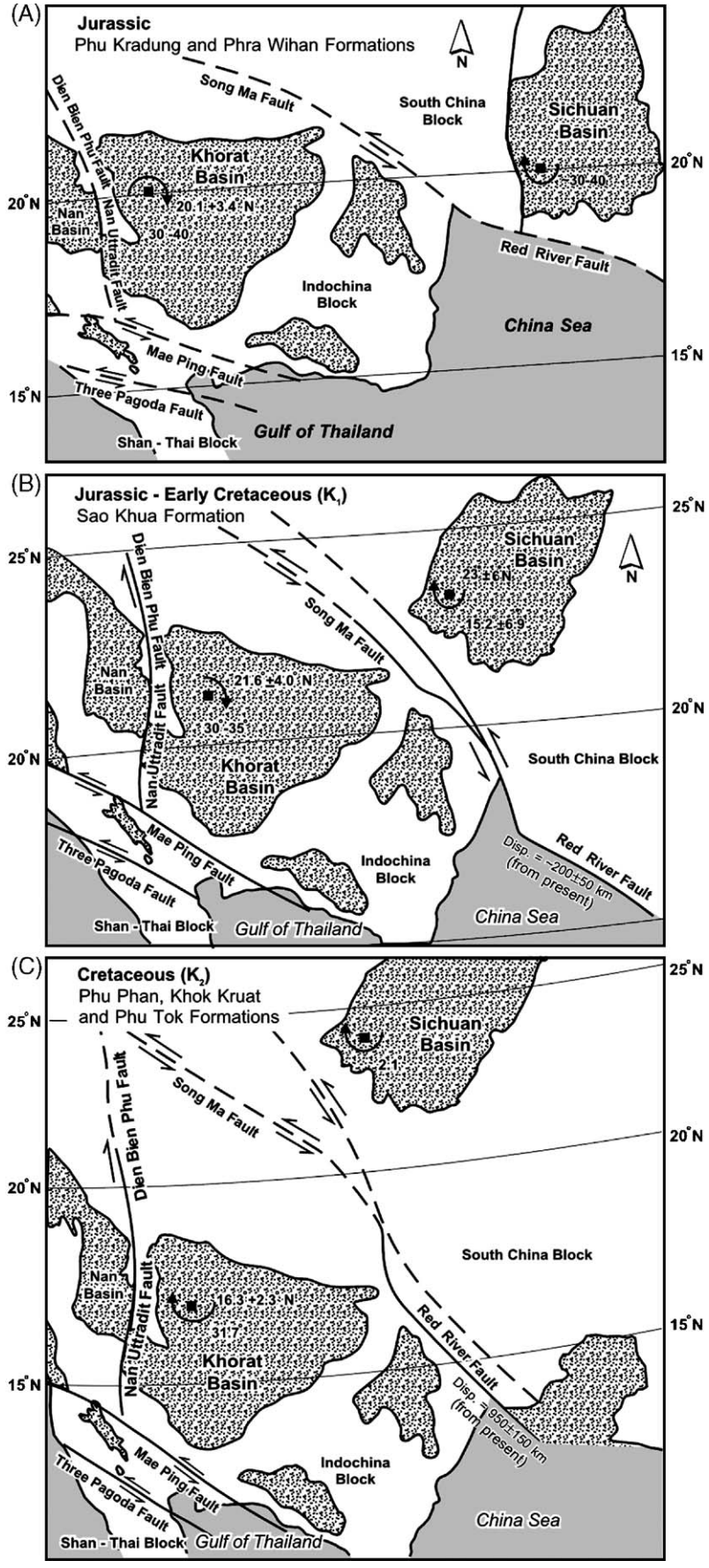
Fig. 6. Paleomagnetic poles and associated 95% confidence circle or dp, dM from Indochina (Upper Triassic to Lower Cretaceous), South China (Upper Triassic to Cretaceous) and this study. Kl_{AP} = Lower Cretaceous APWP Mean South China; Ju_{AP}–Kl_{AP} = Upper Jurassic to Lower Cretaceous APWP Mean South China; Jm_{AP} = Middle Jurassic APWP Mean South China; TRu_{AP} = Upper Triassic APWP Mean South China; Kl_{IN} = Lower Cretaceous Indochina; Ju_{IN}–Kl_{IN} = Upper Jurassic to Lower Cretaceous Indochina; Jm_{IN}–Kl_{IN} = Middle Jurassic to Lower Cretaceous Indochina; Jm_{IN} = Middle Jurassic Indochina; TRu_{IN}–Jl_{IN} = Upper Triassic to Lower Jurassic Indochina; TRu_{IN} = Upper Triassic Indochina; Kl_{SC} = Lower Cretaceous Red bed, South China; Ju_{SC}–Kl_{SC} = Upper Jurassic to Lower Cretaceous red bed, South China; Ju_G = Upper Jurassic Guanyin; and Km_J–Ku_J = Middle to Upper Cretaceous Jiangsu, Hubei.

China block are mostly similar to those of the Jurassic–Cretaceous rocks of NE Thailand. Indochina shows a paleopole position at about 54–65°N, 171–195°E. In contrast, the APWP and VGP Late Jurassic and Cretaceous poles of the stable microplates, north of the Red River Fault including the Yunnan, Qiangtang, and Chuxiong areas exhibit a higher pole-latitude (72.3°N, 208.6°E with A_{95}=13.5°). Enkin et al. (1992) concluded that South China and Indochina Blocks were in contact before Jurassic times and probably were subsequently separated during Late Jurassic to Early Cretaceous times.

South China paleolatitudes during the Early Mesozoic are all at substantially lower latitudes than the present-day latitudes. This difference increases as the paleomagnetic results become older (Enkin et al., 1992; Yang, 1992). It is concluded that the South China and Indochina plates were located at about 20°N of present-latitude at least in Jurassic times, about 10°N in Triassic, and nearly half about 5–6°N in Permian. After that, the northward movement to 25°N of South China probably took place during Late Jurassic along the sinistral Red River Fault. The movement exceeded 950 ±150 km (reference point at 26°N, 102°E) (Yang, 1992) whereas the Yunnan area (south of the Red River Fault) and the Khorat Plateau on Indochina plate also recorded a southward translation during Jurassic–Cretaceous times (Huang and Opdyke, 1993 and in this study).

Jurassic and some Cretaceous paleomagnetic declination data from South China show orientation to the northeast whereas most of the Late Jurassic–Early Cretaceous rocks of

Fig. 7. A model showing the movement of the Indochina and South China plates along the Red River fault zone with a left lateral sense of movement in Mesozoic Period, based mainly upon paleomagnetic data of Yang (1992) and this paper. (A) Early to Late Jurassic. Both plates connected to each other at approximate latitudes between 15° and 20°N, along the zone of weakness (so-called Red River Fault or RRF zone) and with the declination (D) counter-clockwise from the present-day north about 30–40°; (B) Late Jurassic–Earliest Cretaceous. Both plates split from each other, and the South China plate moved with a clockwise rotation about 20–25°, until its declination was 15.2±6.9° counter-clockwise from present-day north. Indochina showed a very small clockwise rotation and moved southward relative to South China; (C) Early–Late Cretaceous. Both plates were displaced from each other along the RRF. South China moved to latitudes 25–26° N and rotated clockwise about 10–15°, until its declination was almost the same as the present-day value. Indochina moved southward and rotated 20–25° clockwise relative to the South China.


(A)
Jurassic
Phu Kradung and Phra Wihan Formations
N
Song Ma Fault
South China Block
Sichuan Basin
Dien Bien Phu Fault
Khorat Basin
20.1 ±3.4° N
30°-40°
Nan Basin
Nan Uttradit Fault
Red River Fault
Indochina Block
China Sea
Mae Ping Fault
Three Pagoda Fault
Gulf of Thailand
Shan - Thai Block
20°N
15°N
(B)
Jurassic - Early Cretaceous (K1)
Sao Khua Formation
25°N
20°N
15°N
Sichuan Basin
23 ±6°N
15.2 ±6.9
Dien Bien Phu Fault
Song Ma Fault
Nan Basin
21.6 ±4.0° N
30°-35°
Nan Uttradit Fault
Khorat Basin
South China Block
Mae Ping Fault
Three Pagoda Fault
Indochina Block
Red River Fault
Disp. = ~200±50 km (from present)
Shan - Thai Block
Gulf of Thailand
China Sea
(C)
Cretaceous (K2)
Phu Phan, Khok Kruat and Phu Tok Formations
Sichuan Basin
Dien Bien Phu Fault
Song Ma Fault
South China Block
Nan Basin
16.3 ±2.3° N
31.7°
Nan Uttradit Fault
Red River Fault
Khorat Basin
Disp. = 950±150 km (from present)
Mae Ping Fault
Three Pagoda Fault
Indochina Block
China Sea
Shan - Thai Block
Gulf of Thailand

the stable South China block has a NNE to N-declination (the Yangtze block — $D=19.1°$ and the South China Plate — $D=15.2\pm6.9°$, Courtillot and Besse, 1986). In contrast, rocks of the Khorat Plateau preserves northeastern declinations between Early Triassic and early Late Cretaceous times. The Yunnan terrane also shows this stability during Jurassic to Cretaceous times (Huang and Opdyke, 1993). Therefore, we interpret that differential orientation of Indochina block relative to South China block was caused by a continuous sinistral movement along the Red River and associated Song Ma faults during the Late Jurassic–Early Cretaceous times. We also conclude that the Khorat Plateau was rotated 14–15° counterclockwise (CW) relative to South China block since this period of times and that the movement and rotation (relative with South China block) are likely to have taken place since Earliest Cretaceous.

8.5. Paleoposition of the Khorat plateau relative to the Shan-Thai block

Shan-Thai Jurassic to recent paleomagnetic poles are also summarized in Table 2. Pole-latitude data for Cretaceous (Jurassic) Kalaw red beds in Myanmar slightly differs from that of the Jurassic–Cretaceous rocks of NE Thailand. In contrast, the paleomagnetic results of Jurassic rocks in western Thailand are quite similar. The mean declination of Cretaceous rocks in the Shan-Thai block is rotated 15.4°CW relative to the Early Cretaceous Khorat Group rocks of the Indochina block. This rotation may be responsible for extension in the Central Thai basin as well as the uplift and unroofing of the Doi Inthanon Complexes in northern Thailand (Richter and Fuller, 1996).

The Shan-Thai paleolatitudes during Cretaceous times moved northwards 12°N relative to the Indochina block (or the Khorat Plateau). This probably took place through the dextral motion of minor plates following the extrusion model of Richter and Fuller (1996).

8.6. Plate movement of the Khorat Group since the Triassic

Since the Late Triassic Period, Thailand and South China have been parts of a single rigid continental block that rotated clockwise to 30–40° (Maranate and Vella, 1986; Enkin et al., 1992, Takemoto et al., 2005). This single united continent was located at latitudes ranging from 10° to 30°N. Later on the Nam Phong Formation began to be deposited in the Early Jurassic as the South China block began to move north to northwestward and collided with the North China block before the Late Jurassic (Yang, 1992). During this time, we believe that Indochina was still together with South China.

Subsequently, in the Late Jurassic to earliest Cretaceous, the Indochina and South China (Yunnan and Chuxiong, Yang, 1992) terranes moved southward by sinistral movement of the Red River (the age between the Sao Khua and Phu Phan Formations). The length of movement is not accurately resolvable by paleomagnetic means, but may be about 950 km based upon the difference in paleolatitudes of earliest Cretaceous and Early to Middle Cretaceous in this study (Fig. 7).

The clockwise rotation of Indochina (D ~31.7°) relative to South China (D ~15.2±6.9°) has values of about 16.0–17.0° since Jurassic–Cretaceous times. These two blocks were then tectonically stable until the end of the Cretaceous. The large 20–25°CW rotation of the Indochina block relative to the present occurred after the Cretaceous and probably result from the collision between India and Asia.

9. Conclusions

Based upon our paleomagnetic and field results in conjunction with the earlier paleomagnetic and related field investigations on Cretaceous clastic rocks, we conclude that:

1. Almost all studied Cretaceous clastics exhibit similar magnetic values with an average declination of 31.7° and inclination of 30.3° at $A_{95}=3.7$;
2. A hiatus within Mesozoic Era took place in Indochina during Early Cretaceous times;
3. Indochina block moved southward relative to South China block, and rotated clockwise during the earliest Cretaceous; and
4. South-eastward translation is estimated to be approximately 950±150 km along the northwest–southeast trending Red River and associated faults after Cretaceous.

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